To follow along in your own workspace, please open the following notebooks:

- `CH7-01-Generating Records`
- `CH7-02-Auto Loader`
- `CH7-03-Feature Engineering Streams`
- `CH7-04-Update Maximum Price Feature Table`
- `CH7-05-Wrapping and Logging the Baseline Model`
- `CH7-06-Wrapping and Logging the Production Model`
- `CH7-07-Model Validation`
- `CH7-08-Batch Inference`
- `CH7-09-Production Batch Inference`
- `CH7-10-Auto Label Loader`
- `mlia utils/transactions funcs.py`
- `production streams.yaml`
- `model retraining n inference workflow.yaml`

We recommend using a multi-node CPU cluster for this project. The first four notebooks (`CH7-01` through `CH7-04`) are nearly identical to their previous versions in *Chapter 6*, but the *Chapter 7* versions all point to the production rather than the development catalog. Table names are parameterized in widgets so they can be set in workflows. Here is a list of the essential notebook-specific changes that have been made to the first four notebooks:

- `CH7-01-Generating Records`: Labels and transactions are now being written to separate folders. The data generation components have also been moved to the `transactions_funcs` file in the utils folder.
- `CH7-02-Auto Loader`: The label column is no longer being added to the transactions table. For production, we ensure that the table being written to has `delta.enableChangeDataFeed = true` before the stream starts. If the table property is set after the stream starts, the stream is interrupted and will require a restart. Lastly, if the table property is never set, Lakehouse Monitoring is negatively impacted and will not support continuous monitoring.
- `CH7-03-Feature Engineering Streams`: Similar to the `CH7-02-Pipeline Auto Loader` notebook, table properties are set before any data is written.
- `CH7-04-Update Maximum Price Feature Table`: The code is cleaned up for a step toward production. Specifically, the feature table is updated rather than created once it exists.